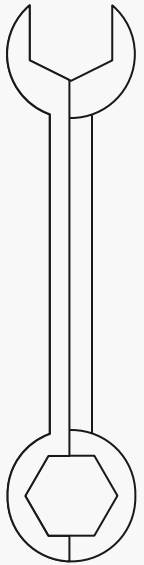
Engineering Design II Project 2: Phase I

Ratchet Wrench

Brett Cavanaugh, Anne Jumper, Ceci LaBarge

LINK TO SLIDESHOW WITH VIDEOS:

https://docs.google.com/presentation/d/1eAgxXTcSMtLEqNSldNwDiy6XwPaHFyRU8-0thEJRLv8/edit?usp=sharing



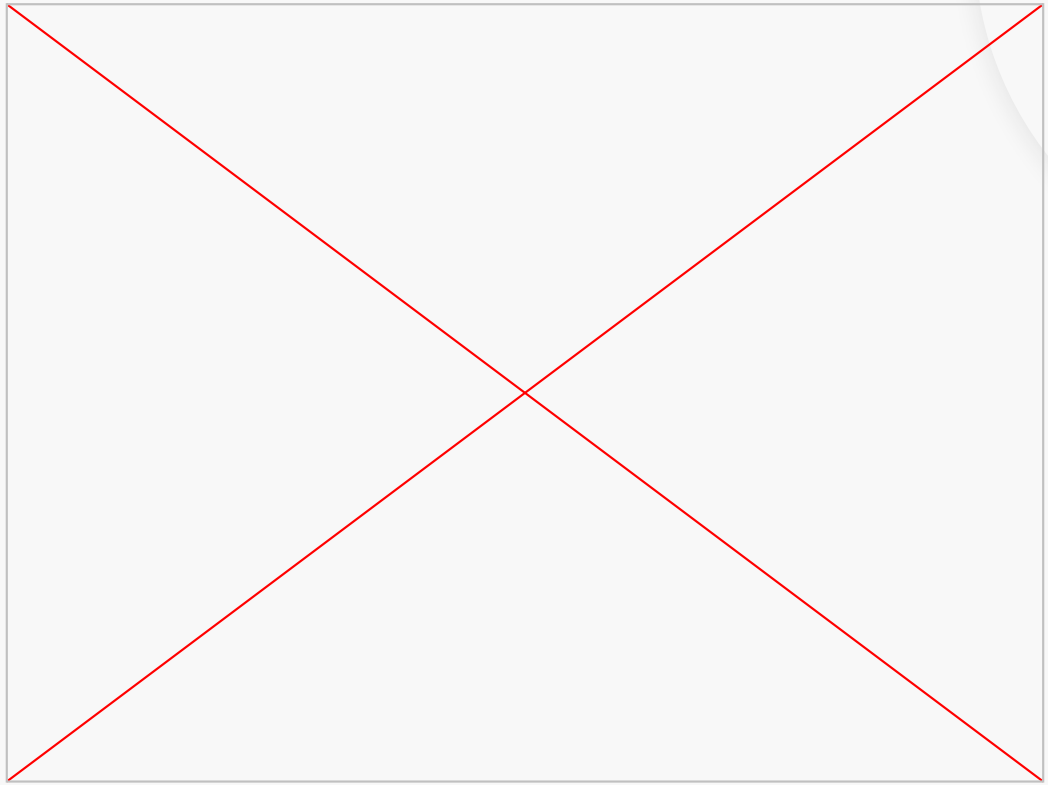
Product Specifications & Images

- 40 ft-lbs max torque
- Weight 6.01 lbs
- Dimensions
  - 5.7” extended head
  - 15.98 x 6.57 x 3.43 inches
- 12V DC power
- Variable speed trigger: 0 - 200 rpm
- Rechargeable lithium ion batteries (x2)
  - Charge capacity over time: 2000 mAh (2.0 Ah)
  - Total energy capacity: 22.2 Wh
- Sockets: 3/8”, 7/16”, 15/32”, 1/2”, 11/16”, 9/16”, 7/16”, 5/8”, 1 x 3/8” x 3” extension rod, 3/8” female to 1/4” male adapter





User Testing

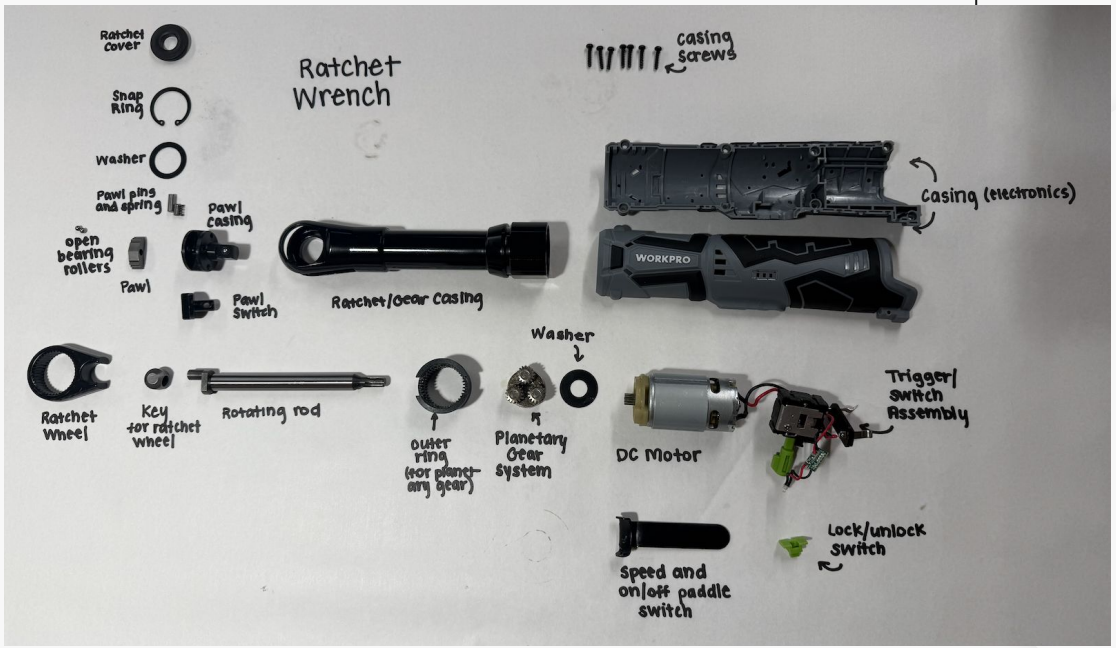


(Video)

Teardown



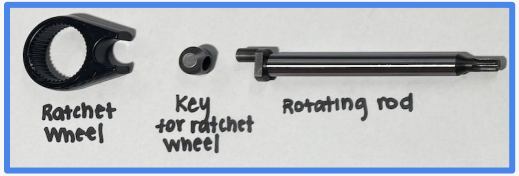
(Video)



Systems & Analyses

Bearings - Brett

- Converts rotary motion to linear motion.
- What loading conditions are expected for this particular system?
- Can the bearing survive fatigue loading over a long period of time?
- Is there a buckling issue?





Planetary Gearbox - Anne

- Reduces speed and increases torque from the motor
- Find the effective gear ratio supplied by the planetary gearbox
- Tooth contact force for max. torque and stress analysis
- How is motor output translated to torque and speed output to the rod?
- How does this affect overall mechanical output?



Ratchet - Ceci

- Analyze the switching of the direction of the pawl
- How does torque affect the effectiveness of the pawl?
- How does the linear motion of the ratchet wheel drive the pawl in a rotary direction?
- Compute the tooth contact force for the max torque (40lb*ft)



